12.2.5.4. ICAC Cross Signing

A joining device, acting also as an Administrator in Fabric B, SHALL obtain the ability to issue NOCs chaining up to the Anchor CA of Fabric A once it has become an Administrator in Fabric A by following the Joint Commissioning Method. For simplicity, the joining device will be called joining Administrator.

To obtain this ability it SHALL receive an ICAC issued by the Anchor Administrator from Fabric A. Cross-signing means that the SubjectPublicKey of the ICAC chaining up to the Anchor CA of Fabric A is identical to the SubjectPublicKey of the ICAC associated with the Fabric indicated by Section 11.25.6.1, “AdministratorFabricIndex” of the joining Administrator (i.e. Fabric B).

A pre-requisite for the ICA Cross Signing process is the execution of FabricFabric Table Vendor ID Verification Procedure against the Fabric indicated by Section 11.25.6.1, “AdministratorFabricIndex” of the joining Administrator. Also the public ICAC of that Fabric SHALL be passed as input parameter to this procedure as `trusted SubjectPublicKey` and it SHALL be encoded using the specific rules for ec-pub-key, pub-key-algo and ec-curve-id.

The ICAC Cross Signing process is started by the Anchor Administrator from Fabric A by sending an Section 11.25.7.1, “ICACCSRRequest” command to the joining Administrator. The joining Administrator SHALL create a Certificate Signing Request (ICA CSR) as described in PKCS #10:

1. ICAC CSR SHALL include a signature (see RFC 2986 section 4.2, SignatureAlgorithm) generated using the Private Key of its pre-installed ICAC.
2. The Public Key associated with the Private Key used to sign the ICA CSR SHALL appear in the SubjectPublicKey of the ICA CSR.
3. Minimum attributes that SHALL be present are shown in the image below.
4. Once the ICAC CSR is created it SHALL be sent in a DER-encoded string inside the ICACCSR parameter of the Section 11.25.7.2, “ICACCSRResponse” command.

ICACSR

```
Certificate Request:
    Data:
        Version: 1 (0x0)
        Subject: ...............
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:12:3b:90:f5:.......
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        Attributes:
        Requested Extensions:
    Signature Algorithm: ecdsa-with-SHA256
```

